

# Project Proposal: AareCast

Oct 2, 2026 · Elena Arganova · Repository: [github.com/nee-source/AareCast](https://github.com/nee-source/AareCast) (public)

## 1 Problem statement

AareCast predicts the Aare water temperature at the BAFU gauge Bern-Schönau 24 h ahead, issued every hour, so swimmers and river-sport clubs in Bern can plan tomorrow's swim.

- **Scope:** one station, year-round; the swim threshold is scored May–September only.
- **Baselines:** persistence (temperature now) and seasonal naive (same hour 24 h ago).
- **Success criterion:** MAE at the 24 h horizon at least 20 % lower than the better baseline on the out-of-time test period. Secondary: precision and recall of "swimmable ( $\geq 18$  °C) tomorrow at 18:00".

## 2 Originality & motivation

I live in Bern and like to swim in the Aare on summer evenings, so "will it be warm enough tomorrow evening?" is a question I ask myself every day. The popular [Aare.guru](#) app answers it only 2 h ahead.

- **Prior work:** none of the 15 [HSLU FS26 projects](#) covers water. KTH ID2223 2025/26 has "[Water Temperature at Södertälje](#)" (a Swedish bathing site); earlier KTH years have river-flood and stream-flow projects.
- **Difference:** AareCast forecasts a lake-fed Swiss river: it uses the upstream Thun gauge (Lake Thun outflow) plus issued MeteoSwiss forecasts, and is benchmarked against a public app's existing 2 h horizon.

## 3 Data source & features

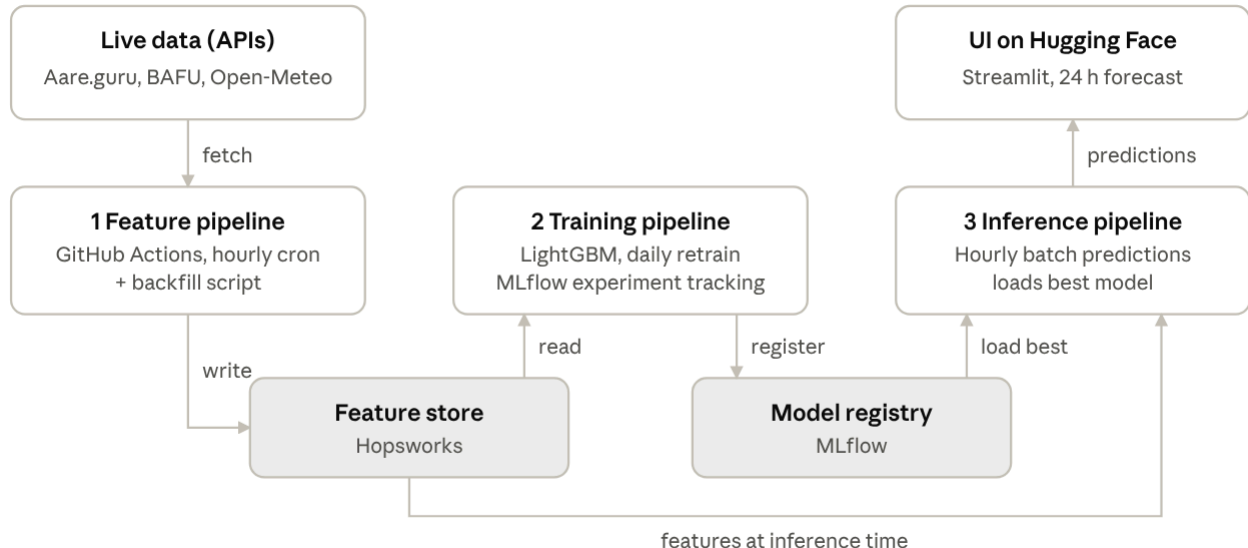
All sources are public APIs (no scraping), free for non-commercial use; Aare.guru and BAFU are credited in the README.

- **Live:** [Aare.guru API](#): Bern water temp, flow, height (BAFU data), every 10 min.
- **Upstream + backfill:** [existenz.ch hydro API](#): BAFU gauges incl. Thun, 10 min, last 32 days; from the first poll the store grows by 24 hourly rows per day. Older years via the existenz.ch archive or a BAFU data request; [hydrodaten.admin.ch](#) is the fallback for all hydro data.

- **Weather:** Open-Meteo (MeteoSwiss ICON-CH): hourly forecasts plus an archive of past forecasts.
- **Label:** measured water temperature at Bern-Schönau at  $t+24$  h. It is a future measurement, not a rule applied to the inputs.
- **Features:** water temp now; lags 1 h, 3 h, 24 h; 24 h change; flow and height at Bern; water temp and flow at Thun (lake outflow, upstream); air temp now; forecast air temp mean/max, sunshine hours and precipitation for the next 24 h; hour-of-day and day-of-year (sin/cos).
- **Rare class:** "swimmable" only occurs in summer, so it is scored on May–September with precision and recall.
- **Split & leakage:** out-of-time split (train on older weeks, validate and test on the most recent ones, no shuffling). Weather features use only forecasts *issued before*  $t$ , never observed future weather. Data is resampled to a fixed 1 h grid before lags are computed, so missed polls cannot shift a lag. Backfilled forecasts from Open-Meteo's archive may flatter the model, so the final test period uses only live-logged forecasts.

## 4 System design

Three automated pipelines share one feature store and one model registry



AareCast FTI architecture · 3 pipelines, feature store, model registry

- **Stack:** GitHub Actions cron (free, in the repo) · Hopworks feature store (same features for training and inference) · MLflow on DagsHub (runs + registry in one) · LightGBM · Streamlit on HF Spaces (free cloud deploy) · Docker, uv.lock, pytest in CI. Optional later: Evidently, Airflow.